Supporting Information:

Supporting Information: Target-Anchored Computational Prioritization of African-Natural-Product-Inspired Antimalarial Candidates

Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou, Wilfred Fon Mbacham, and Serge Guy Nana Engo

E-mail:

This Supporting Information documents the data, pocket definitions, docking settings, quality-control rules, and complete score matrix underlying the main article. It is organized as reproducibility material rather than as a second narrative article. The four-target panel is computational; its scores are not experimental affinities.

S1. Study design and cohort definition

The upstream hybrid library contained 65 856 unique molecules derived from 396 African natural products and 454 synthetic antimalarial compounds. The present target-anchored analysis used the locked 17-member Set-C polypharmacology-oriented cohort. Set A (MPO top-20) and Set B (parent-study MD top-20) were not substituted for Set C. Canonical SMILES, candidate identifiers, source hashes, and inclusion labels are provided in `results/integrated_candidate_`

S2. Target and anchor evidence

Table S1: Target-specific pocket anchors and evidence classes.

Target	PDB	Anchor	Box (Å)	Evidence class
PfDHFR	7F3Y	MTX A702	18	Catalytic-site ligand anchor
PfCRT	6UKJ	Y01 A501	28	Membrane/cavity proxy; not an inhibitor
PfClpP	2F6I	Ser252/His223/Asp219	28	Catalytic-triad anchor
PfATP4	9N10	D451/DPPR	25	Conserved machinery; no inhibitor anchor

The historical 4GM2 structure is PfClpR and is excluded from PfClpP interpretation. The PfCRT and PfATP4 evidence classes are intentionally weaker than a co-crystallized small-molecule pocket and should be treated as exploratory.

S3. Ligand preparation and Vina settings

SMILES were standardized, embedded in three dimensions, and converted to PDBQT with Meeko, following established Vina preparation conventions.[S1,S2] Each target–candidate pair used a fixed receptor and target-specific configuration. No pose was translated after docking, and no search box was expanded after inspecting a result. The geometric gate required a rank-1 centroid in the box, a minimum anchor distance no greater than 10 Å, and an in-box atom fraction of at least 90 %. The full configuration and output hashes are stored with the raw result records. Because the four receptors have different pocket architectures and evidence classes, no arithmetic mean across targets is reported or used for prioritization.

Table S2: Raw Vina scoring-function estimates for the 17-member cohort. Units are kcal mol^{-1}; values are not measured free energies.

Candidate	PfDHFR	PfCRT	PfClpP	PfATP4
PP-01	−5.947	−6.483	−5.467	−6.361
PP-02	−5.562	−6.124	−5.907	−6.591
PP-03	−6.089	−6.633	−6.358	−7.311
PP-04	−5.493	−5.329	−5.386	−6.086
PP-05	−6.285	−6.412	−6.466	−6.378
PP-06	−6.267	−7.907	−7.031	−7.285
PP-07	−6.226	−6.633	−6.226	−5.775
PP-08	−4.935	−5.213	−5.052	−5.494
PP-09	−5.693	−6.871	−5.613	−6.078
PP-10	−6.346	−6.395	−6.016	−6.670
PP-11	−6.179	−6.376	−6.583	−6.724
PP-12	−5.248	−6.357	−5.386	−4.635
PP-13	−5.921	−5.685	−6.454	−7.565
PP-14	−4.863	−5.328	−5.047	−5.460
PP-15	−6.045	−6.084	−6.366	−7.016
PP-16	−5.469	−6.202	−6.203	−6.451
PP-17	−5.266	−5.116	−5.342	−5.360

S4. Complete four-target score matrix

S5. Per-pair geometric quality control

All 68 pairs passed the computational gate. Anchor-distance ranges were 2.00 Å to 3.79 Å for PfDHFR, 3.03 Å to 4.08 Å for PfCRT, 3.29 Å to 8.88 Å for PfClpP, and 1.98 Å to 3.79 Å for PfATP4. These values describe pose placement relative to the declared anchors; they do not measure residence time, binding kinetics, or biological engagement.

S6. What is and is not inferred

The target-anchored panel supports a reproducible computational ranking layer. It does not support claims of confirmed inhibitors, measured IC_{50}/EC_{50}, resistance circumvention, or equivalent pocket validity across targets. Raw hashes and automated integrity checks document reproducibility; they do not substitute for structural or experimental validation.

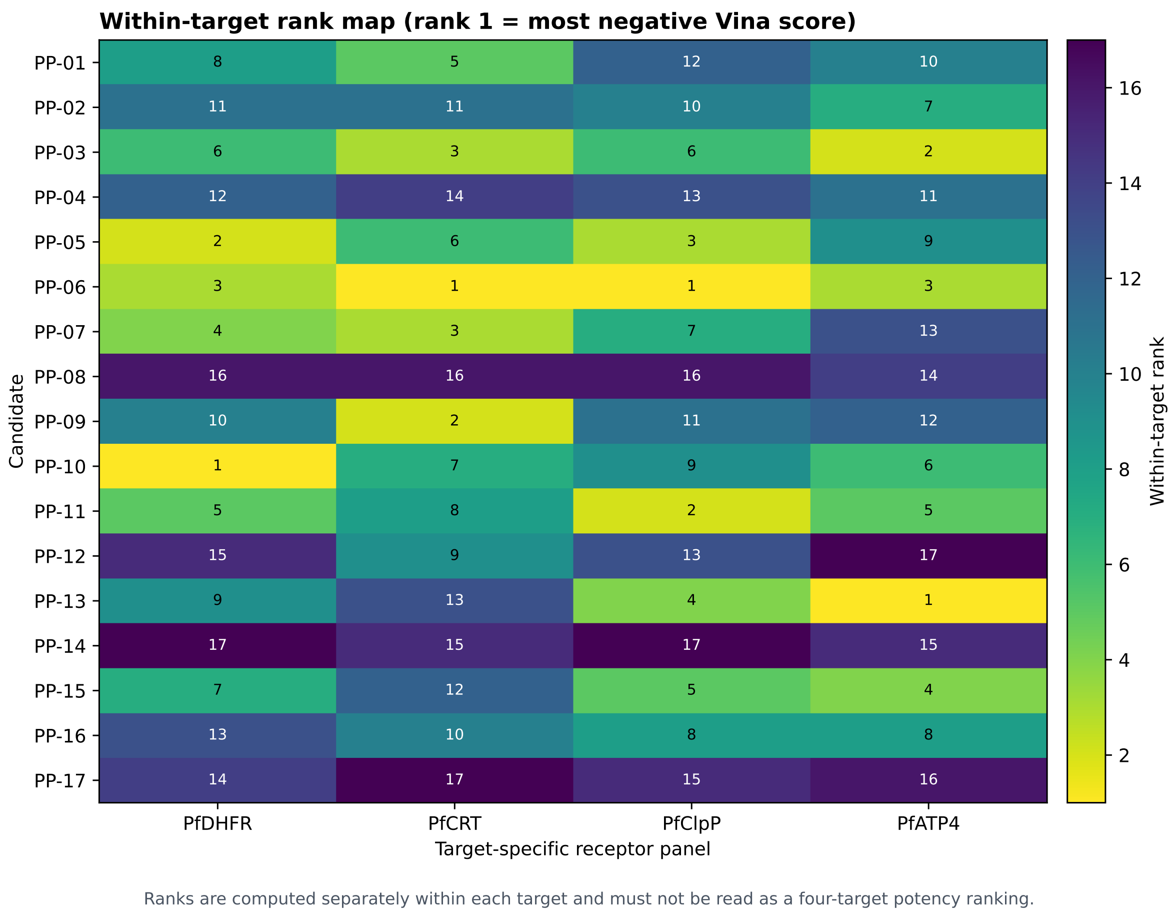

Figure S1: Within-target rank map for the 17 candidates. Rank 1 denotes the most negative Vina score within a target; ranks are calculated independently for each receptor and must not be interpreted as a cross-target potency order. These checks assess pose placement and provenance; they are not biological validation.

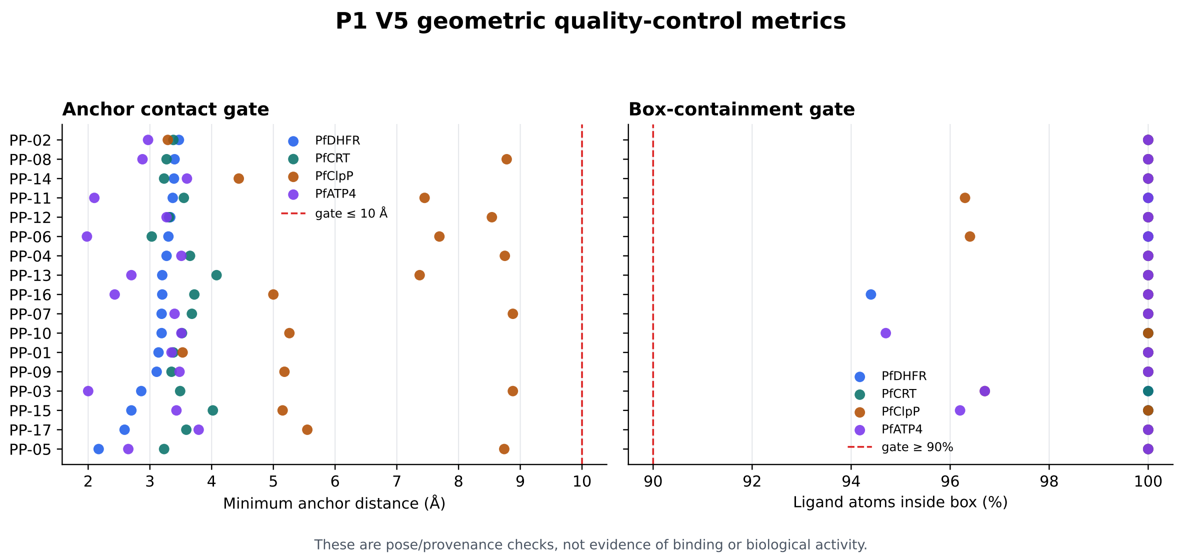

Figure S2: Geometric quality-control metrics for the 68 candidate–target poses. Dashed lines mark the predeclared anchor-distance and in-box-fraction thresholds. These checks assess pose placement and provenance, not binding or biological activity. These checks assess pose placement and provenance; they are not biological validation.

S7. Data and code availability

Raw poses, receptor and ligand files, configuration files, candidate manifests, and analysis scripts are available at `https://github.com/NanaEngo/Malaria_codesV2`. The four-target machine-readable table is `results/v5_four_target_vina_affinities.csv`; the per-pair review table is `results/v5_four_target_vina_review_table.json`.

References

(S1) Trott, O.; Olson, A. J. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry* **2009**, *31*, 455–461.

(S2) Eberhardt, J.; Santos-Martins, D.; Tillack, A. F.; Forli, S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling* **2021**, *61*, 3891–3898.